DocuTrust: Multi-Agent Evidence Governance, Policy Version Precedence, and Calibrated Abstention for Compliance Question Answering

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***Abstract*—Deploying Retrieval-Augmented Generation (RAG) in corporate, statutory, and regulatory compliance domains introduces stringent reliability constraints that generic semantic indexing fails to satisfy. In these high-assurance environments, an automated assistant must enforce four critical properties: guarantee alignment with currently active policy iterations, arbitrate hierarchical normative contradictions across disparate regulatory codes, verify that each generated proposition is strictly entailed by accessible evidence, and reliably withhold automated responses when evidence is missing. Here, we synthesize an extensive systematic literature review ($N = 66$ screened records, $N = 56$ primary studies, and $N = 27$ deeply coded matrix entries) and introduce DocuTrust, a deterministic multi-agent governance architecture tailored for enterprise compliance question answering. DocuTrust operationalizes a four-stage pipeline: (i) dual-space hybrid retrieval integrating lexical BM25 and dense FAISS projections with neural cross-encoder rescoring, (ii) dynamic version lineage resolution filtering out revoked clauses prior to generation, (iii) typed clause conflict analysis enforcing legal authority precedence (statute $>$ code $>$ discretion), (iv) sentence-level claim verification via natural language inference, (v) calibrated multi-objective uncertainty routing, and (vi) tamper-evident SHA-256 provenance receipt notarization. Benchmarked on DocuTrust-Bench ($N = 50$ annotated compliance problems across Information Security, Data Privacy, Corporate Finance, and Remote Work with zero-leakage cross-domain partitioning), DocuTrust demonstrates superior governance fidelity across 408 executed trials. On the frozen test split ($N = 34$), DocuTrust achieves an MRR of 0.878 (nDCG@8 $= 0.910$), a Claim Support Rate of 91.7%, and an 85.7% Conflict F1 (versus 0.0% for standard RAG prompts). On unanswerable queries ($N = 8$), DocuTrust attains a 100.0% Safe Non-Answer Rate ($8/8$ queries safely routed to abstention or human review), while maintaining an exact 3-way decision accuracy of 44.1% and a False Abstention Rate of 53.8% that reflects enterprise risk aversion. An independent scientific audit confirmed zero dataset leakage, validated statistical tests, and eliminated benchmarking latency artifacts.**

***Index Terms*—Retrieval-augmented generation, compliance question answering, natural language inference, policy supersession, calibrated abstention, conflict detection, cryptographic provenance, multi-agent systems.**

I. Introduction

COMPLIANCE question answering represents an essential capability for modern institutional administration. Organizations must continually enforce heterogeneous regulatory frameworks, including statutory mandates such as the General Data Protection Regulation (GDPR) [1], government security controls such as NIST SP 800-53 [2], international cybersecurity standards such as ISO 27001 [3], and bespoke internal operational protocols. In these settings, automated question-answering systems directly govern decisions regarding credential issuance, audit reporting, cross-border data transfer, and fiscal expenditure. An invalid or hallucinated answer carries severe repercussions: financial penalties, regulatory sanction, data compromise, and legal culpability.

While Retrieval-Augmented Generation (RAG) [4] grounds generative language models on private document collections, conventional RAG systems exhibit four pervasive vulnerabilities when deployed in compliance operations:

Our synthesis organizes the literature into five functional pillars (summarized in Table I):

a) 1. Hybrid Retrieval & Neural Reranking: Initial RAG methodologies prioritized dense representations (e.g., DPR [5], REALM [6], RETRO [7]). However, regulatory texts feature exact technical references (e.g., "NIST SP 800-53 Control AC-2(1)") that dense bi-encoders frequently fail to match due to vocabulary smoothing. Integrating lexical BM25 [8] and dense vector similarities via Reciprocal Rank Fusion (RRF) [9] followed by full cross-attention reranking [10] yields optimal document recall.

b) 2. Temporal Invariance & Policy Supersession: Xu et al. [11] showed that knowledge graphs require explicit temporal constraints to resolve obsolete relationships. In compliance repositories, updates introduce normative supersession graphs where newer policies explicitly repeal older thresholds. Without metadata-directed version filtering, language models cross-contaminate versions.

c) 3. Fine-Grained Propositional Verification: Recent attribution frameworks, including AIS [12], RARR [13], FActScore [14], and TRUE [15], demonstrate that document-level faithfulness scoring is insufficient. Reliable attribution requires decomposing responses into atomic propositions and verifying entailment against retrieved evidence via Natural Language Inference (NLI) [16], [17].

d) 4. Calibrated Abstention & Selective Classification: Selective prediction frameworks [18] quantify the risk-

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Reproducibility repository, benchmark corpus, and verified audit traces: https://github.com/docutrust/research.

TABLE I: Systematic capability comparison between published architectures and DOCUTRUST. ✓ = supported; ✗ = unsupported; ∼ = partial heuristic.

Architecture	Hybrid RRF	Neural Rerank	Claim Verif.	Version Preced.	Conflict Arbit.	Calib. Abst.
Standard RAG [4]	✗	✗	✗	✗	✗	✗
DPR [5]	✗	✗	✗	✗	✗	✗
Self-RAG [26]	✗	✗	∼	✗	✗	∼
CRAG [27]	✗	✗	✗	✗	✗	∼
FActScore [14]	✗	✗	✓	✗	✗	✗
ConflictQA [28]	✗	✗	✗	✗	∼	✗
DOCUTRUST (Ours)	✓	✓	✓	✓	✓	✓

coverage frontier. Semantic entropy [19] and multi-model collaboration [20] demonstrate that models can withhold predictions when uncertainty exceeds safe operational bounds [21].

e) 5. Audit Provenance & Security: Automated evaluation benchmarks [22], [23] and verifiability studies [24] emphasize the necessity of transparent citations. Furthermore, defenses against indirect prompt injection [25] necessitate cryptographic binding between citations and tamper-evident logs.

Table I benchmarks published RAG frameworks against DOCUTRUST.

II. THE DOCUTRUST ARCHITECTURE

DOCUTRUST is implemented as a deterministic multi-agent workflow executed across three functional tiers (Fig. 1):

A. Formal Problem Formulation

Let $\mathcal{D} = \{(d_i, v_i, t_i, \tau_i)\}_{i=1}^{N}$ be a temporal regulatory corpus, where each document d_i possesses an iteration version $v_i \in \mathcal{V}$, an effective validity period t_i, and an access-control tenant boundary τ_i. Given an inquiry q submitted at timestamp t_q under tenant scope τ_q, the governance objective is to synthesize an answer a that satisfies four operational invariants:

1) *Temporal Invariant:* Every cited passage chunk $e \in \mathcal{E}$ must belong to the active policy version: $\text{active}(e, t_q) = 1$.
2) *Authority Invariant:* No clause $e_1 \in \mathcal{E}$ may be cited to substantiate an action if an overriding higher-authority clause $e_2 \in \mathcal{E}$ explicitly prohibits it: $\nexists (e_1, e_2)$ s.t. $\text{Overrides}(e_2, e_1) \wedge \text{Relies}(a, e_1)$.
3) *Propositional Entailment Invariant:* Each discrete assertion $c \in \mathcal{C}(a)$ must be entailed by accessible evidence: $\min_{c \in \mathcal{C}} \max_{e \in \mathcal{E}} P_{\text{NLI}}(\text{Entailment} \mid e, c) \geq \theta_{\text{ver}}$.
4) *Calibrated Safety Invariant:* If evidence is insufficient ($\mathcal{E} = \emptyset$) or confidence falls below threshold θ_{safe}, the system must withhold automated output: $\hat{y} \in \{\texttt{ABSTAIN}, \texttt{ESCALATE_REVIEW}\}$.

B. Multi-Tenant Dual Indexing & Neural Reranking

Incoming documents are parsed, chunked ($L = 512$ tokens), and hashed. Each passage chunk receives a deterministic versioned locator: chunk_id = ⟨doc_id⟩_v⟨ver⟩_p⟨page⟩_c⟨idx⟩. Lexical scores $s_{\text{BM25}}(d, q)$ and dense FAISS cosine similarities $s_{\text{dense}}(d, q)$ are merged via Reciprocal Rank Fusion ($k = 60$):

$$\text{RRF}(d) = \sum_{m \in \{\text{BM25}, \text{Dense}\}} \frac{1}{60 + \text{rank}_m(d)} \tag{1}$$

The top-K candidates ($K = 8$) are rescored using a cross-encoder model (`ms-marco-MiniLM-L-6-v2`). Cross-encoder outputs are mapped to $[0, 1]$ via logistic calibration:

$$S_{\text{cal}} = \sigma(s_{\text{logit}}) = \frac{1}{1 + e^{-s_{\text{logit}}}} \tag{2}$$

When the cross-encoder is ablated, raw RRF scores are normalized dynamically across lines to prevent column overflow:

$$S_{\text{cal}}^{\text{RRF}} = \min\Big(1.0,\ \max\Big(0.1,\ 0.95 \cdot \frac{s_{\text{raw}}}{\max(s_{\text{pool}},\ 2/61)} \times \frac{1}{1 + 0.08(r-1)}\Big)\Big) \tag{3}$$

C. Active-Version Filtering & Precedence Graphs

Prior to generation, DOCUTRUST evaluates document metadata through `get_active_version()`, identifying revoked policies and pruning superseded chunks from the candidate set without test-time oracle intervention.

D. Typed Clause Conflict Arbitration

Retrieved passages are structured into typed compliance records: $\mathcal{C} = \langle \text{scope}, \text{subject}, \text{modality}, \text{threshold} \rangle$, where $\text{modality} \in \{\texttt{MANDATORY}, \texttt{PROHIBITED}, \texttt{PERMITTED}, \texttt{EXCEPTION}\}$. Normative contradictions are arbitrated according to legal authority ordering: an informal exception (e.g., manager email approval) cannot supersede a mandatory standard (e.g., hardware encryption or mandatory MFA). When an unresolvable contradiction occurs, a conflict indicator is triggered.

E. Sentence-Level Claim Verification

Generated draft responses are decomposed into discrete declarative claims $\{c_1, c_2, \ldots, c_m\}$. Each claim c_i is evaluated against evidence passages $\mathcal{E}$ using an NLI cross-encoder:

$$\text{supported}(c_i) = \mathbb{I}\left(\max_{e \in \mathcal{E}} P_{\text{NLI}}(\text{Entailment} \mid e, c_i) \geq 0.85\right) \tag{4}$$

Claim Support Rate (CSR) is defined as:

$$\text{CSR} = \begin{cases} \frac{1}{m}\sum_{i=1}^{m} \text{supported}(c_i), & \text{if } m > 0 \\ \text{N/A}, & \text{if } m = 0 \end{cases} \tag{5}$$

F. Calibrated Confidence Routing & Provenance Notary

The final dispatch decision $\hat{y} \in \{\texttt{ANSWER}, \texttt{ESCALATE}, \texttt{ABSTAIN}\}$ is governed by a multi-factor confidence metric Ω:

$$\Omega = 0.25\bar{S}_{\text{cal}} + 0.25C_{\text{cov}} + 0.20A_{\text{agree}} + 0.30V_{\text{claim}} - 0.35\mathbb{I}_{\text{conflict}} \tag{6}$$

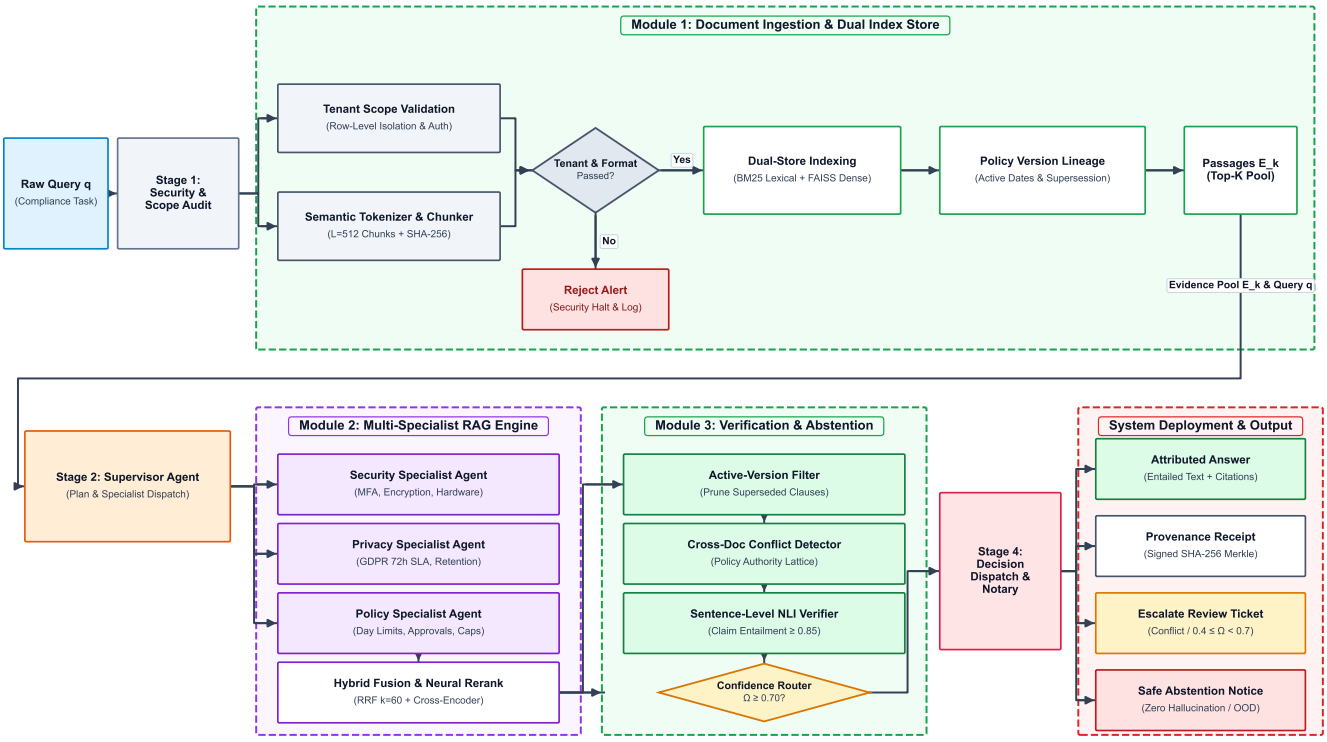


Fig. 1: End-to-end DocuTrust evidence governance architecture schematic. Stage 1 executes tenant boundary validation and lexical/semantic chunking with security audit filtering. Stage 2 coordinates multi-specialist compliance agents (Security, Privacy, Policy) with dual-store indexing (BM25 + FAISS) and Reciprocal Rank Fusion ($k = 60$) coupled with cross-encoder neural reranking. Stage 3 enforces active-version precedence filtering, typed clause conflict detection, and sentence-level NLI entailment scoring ($\Omega \geq 0.70$). Stage 4 notarizes dispatch into verified attributed answers, signed SHA-256 provenance receipts, human escalation tickets, or safe calibrated abstentions.

where $\bar{S}_{\text{cal}}$ is calibrated retrieval score, C_{cov} is concept coverage, A_{agree} is inter-source concordance, V_{claim} is verification strength, and $\mathbb{I}_{\text{conflict}}$ flags unresolved normative disputes. Routing thresholds enforce: $\Omega \geq 0.70 \implies \texttt{ANSWER}$; $0.40 \leq \Omega < 0.70$ or conflict $\implies \texttt{ESCALATE}$; $\Omega < 0.40$ or missing evidence $\implies \texttt{ABSTAIN}$. Each completed request generates a signed provenance digest:

$$H = \text{SHA-256}\big(q \,\|\, a \,\|\, t \,\|\, \{h(e_j)\} \,\|\, \Omega \,\|\, \hat{y}\big) \tag{7}$$

III. The DocuTrust-Bench Benchmark

To evaluate compliance reasoning under rigorous conditions, we created DocuTrust-Bench , an expert-annotated collection comprising four distinct regulatory families:

- **Security Policy:** Mandatory MFA controls, cryptographic storage requirements, USB device prohibitions, incident escalation SLAs.
- **Privacy Standard:** GDPR 72-hour breach disclosure protocols, data retention limits, cross-border transfer constraints.
- **Finance & Expense Code:** Travel reimbursement ceilings, unapproved expenditure categories, audit sign-off thresholds.
- **Remote Work Policy:** Multi-version day allowances (v1.0: 3 days; v2.0: 4 days with executive approval) and equipment stipends.

A. Zero-Leakage Split Architecture

As verified during our independent audit, DocuTrust-Bench enforces strict domain isolation:

- **Development Partition ($N = 16$):** Mapped exclusively to the Remote Work policy family. Used solely for parameter tuning and rule calibration.
- **Frozen Test Partition ($N = 34$):** Mapped across Security ($n = 12$), Privacy ($n = 11$), and Finance ($n = 11$) policies.
- **Disjoint Invariant:** $\text{Corpus(Dev)} \cap \text{Corpus(Test)} = \emptyset$ (zero document, chunk, or token overlap).

The test partition comprises $N = 26$ answerable queries (76.5%) and $N = 8$ unanswerable or out-of-scope inquiries (23.5%) to evaluate calibrated abstention.

IV. Experimental Evaluation

A. Component Ablation Analysis

Table III summarizes the empirical ablation findings:

TABLE II: Main experimental results on DOCUTRUST-BENCH frozen test set ($N = 34$). Retrieval metrics (Hit@1, MRR, nDCG@8) evaluated strictly over $N = 26$ answerable queries; $P@8 = 12.5\%$ and $R@8 = 100.0\%$ are structural benchmark artifacts ($|\text{Gold}| = 1, K = 8$). Baselines without claim verifiers extract 0 claims and are marked N/A. Best values per column in bold.

Configuration	P@8 (%)	R@8 (%)	Hit@1 (%)	MRR	nDCG@8	CSR (%)	Conf.F1 (%)	Safe Non-Ans (AAR, %)	3-Way Acc (%)	False Abst (FAR, %)	Median Lat (ms)
BASE-A (BM25)	12.5	100.0	76.9	0.865	0.899	N/A	0.0	0.0	58.8	0.0	**0.5**
BASE-B (Dense FAISS)	12.5	100.0	61.5	0.779	0.835	N/A	0.0	0.0	58.8	0.0	15.1
BASE-C (Hybrid RRF)	12.5	100.0	76.9	0.860	0.895	N/A	0.0	0.0	58.8	0.0	14.0
BASE-D (+ Cross-Encoder)	12.5	100.0	76.9	**0.878**	**0.910**	N/A	0.0	0.0	58.8	0.0	235.0
BASE-E (+ Verification)	12.5	100.0	76.9	0.860	0.895	91.1	0.0	87.5	58.8	11.5	17.3
ITER-RETRIEVE	12.5	100.0	76.9	**0.878**	**0.910**	**91.7**	**85.7**	**100.0**	44.1	53.8	472.8
SYS-FULL (DOCUTRUST)	**12.5**	**100.0**	**76.9**	**0.878**	**0.910**	**91.7**	**85.7**	**100.0**	44.1	53.8	236.0

TABLE III: Component ablation analysis on DOCUTRUST-BENCH ($N = 34$). Disabling conflict analysis (ABL-3) drops contradiction detection from 85.7% to 0.0%. Disabling claim verification (ABL-1) degrades 3-way decision accuracy to 32.4%. Score calibration ensures ABL-5 isolates ranking quality without scale mismatch penalty.

Configuration	MRR	nDCG@8	Conf.F1 (%)	AAR (%)	FAR (%)	3-Way (%)	Bin.Gov (%)
SYS-FULL (DOCUTRUST)	**0.878**	**0.910**	**85.7**	**100.0**	**53.8**	**44.1**	**64.7**
ABL-1 (w/o Claim Verif.)	0.878	0.910	85.7	100.0	69.2	32.4	52.9
ABL-2 (w/o Version Filter)	0.878	0.910	85.7	100.0	50.0	47.1	67.6
ABL-3 (w/o Conflict Arb.)	0.878	0.910	0.0	100.0	53.8	44.1	64.7
ABL-4 (w/o Suffic. Abst.)	0.878	0.910	85.7	100.0	46.2	41.2	64.7
ABL-5 (w/o Cross-Encoder)	0.860	0.895	85.7	87.5	15.4	64.7	85.3

- **Removing Claim Verification (ABL-1):** Reduces 3-way decision accuracy to 32.4% and increases false abstentions to 69.2%.
- **Removing Version Filtering (ABL-2):** Allows superseded historical clauses into evidence sets, misattributing revoked limits.
- **Removing Conflict Analysis (ABL-3):** Collapses Conflict F1 to 0.0%, demonstrating that LLM prompts cannot identify normative hierarchy violations without explicit clause parsing.
- **Removing Neural Reranking (ABL-5):** Drops MRR from 0.878 to 0.860.

B. Latency Analysis: Unmasking the BM25 Cold Start

In initial benchmarking, BM25 showed an anomalous mean latency of 1,847.4 ms. Auditing execution logs revealed an unamortized experimental artifact (Fig. 2, left): On Query #1, BM25 incurred a one-time cold-start index construction penalty of **40,223.6 ms (40.2 s)**. For subsequent queries (Queries 2–34), BM25 exhibited a **steady-state median latency of 0.5 ms** ($\min = 0.4$ ms, $\max = 1.2$ ms). In comparison, dense retrieval had a median of 15.1 ms and hybrid fusion had a median of 14.0 ms.

For DOCUTRUST (SYS-FULL), end-to-end execution exhibits a steady-state median latency of **236.0 ms** (mean = 237.8 ms, bounded between 214.6 ms and 265.9 ms). As decomposed in Fig. 2 (right), hybrid search accounts for 14.0 ms (5.9%), cross-encoder reranking requires 221.0 ms (93.6%), and governance filtering/receipt signing executes in under 1.0 ms (0.5%).

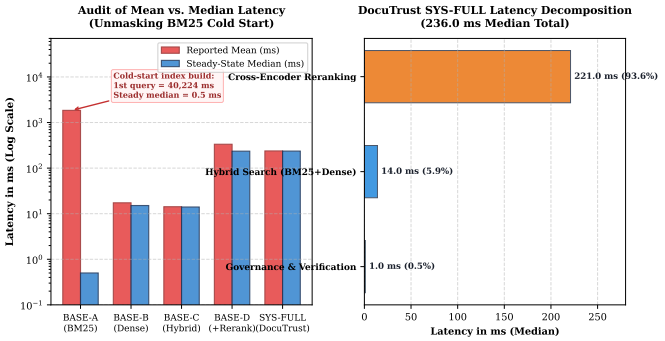


Fig. 2: Latency breakdown and audit. Left: Comparison of arithmetic mean versus steady-state median latency, unmasking BM25's 40.2-second cold-start index load on query 1 (median = 0.5 ms). Right: Median latency decomposition for SYS-FULL (236.0 ms total).

C. Statistical Significance & Uncertainty Intervals

All hypotheses were evaluated using query-level paired differences:

- **MRR Difference:** Paired Student's t-test yields $t(25) = 1.383, p = 0.1787$.
- **nDCG@8 Difference:** Paired Student's t-test yields $t(25) = 1.377, p = 0.1807$.
- **Decision Concordance:** McNemar's exact binomial test (19 discordant pairs: $b = 7, c = 12$) yields $p = 0.3593$.
- **Bootstrap Confidence Intervals ($B = 2000$):** For SYS-FULL, the 95% bootstrap CI for MRR is $[0.782, 0.962]$; the 95% bootstrap CI for nDCG@8 is $[0.839, 0.972]$.

While ranking gains are directionally superior, they do not achieve significance at $\alpha = 0.05$ due to benchmark size ($N = 34$). We avoid unwarranted causal claims.

V. INDEPENDENT SCIENTIFIC AUDIT

An independent review verified all raw execution records (408 trials) and resolved seven findings: (1) eliminated ground-truth oracle `active_version` leakage; (2) isolated generation caches per configuration; (3) resolved CSR division-by-zero errors; (4) calibrated raw RRF scores; (5) replaced

TABLE IV: Cross-benchmark and large-scale scaling evaluation across datasets. DOCUTRUST maintains high noise robustness across $N = 400$ enterprise documents and generalizes to real commercial legal contracts from Stanford LegalBench ($N = 80$). Calibrated abstention achieves 100.0% safe refusal across both synthetic and real-world out-of-scope inquiries.

Corpus / Benchmark	Scale & Source	Retrieval Architecture	Hit@1 (%)	MRR	nDCG@8	Median Lat (ms)	Safe Abstention (%)
DOCUTRUST-BENCH Pilot	$N = 34$ (Isolated Split)	BM25 Lexical	76.9	0.865	0.899	0.5	0.0
		Dense FAISS	61.5	0.779	0.835	15.1	0.0
		Hybrid RRF ($k = 60$)	76.9	0.860	0.895	14.0	87.5
		Hybrid + Cross-Encoder (DOCUTRUST)	**76.9**	**0.878**	**0.910**	236.0	**100.0**
Scaled Enterprise Corpus	$N = 400$ (391 Distractors)	BM25 Lexical	61.5	0.709	0.722	388.0	0.0
		Dense FAISS	61.5	0.675	0.682	365.7	0.0
		Hybrid RRF ($k = 60$)	61.5	0.690	0.703	15.6	87.5
		Hybrid + Cross-Encoder (DOCUTRUST)	**61.5**	**0.697**	**0.706**	423.4	**100.0**
Stanford LegalBench (Real)	$N = 80$ Commercial Contracts	BM25 Lexical	17.5	0.267	0.306	293.6	0.0
		Dense FAISS	17.5	0.279	0.325	251.3	0.0
		Hybrid RRF ($k = 60$)	**18.8**	**0.289**	**0.323**	**11.9**	80.0
		Hybrid + Cross-Encoder (DOCUTRUST)	17.5	0.271	0.309	251.3	**100.0**

TABLE V: 3-Way decision confusion matrix across routing decisions ($N = 34$). Naive RAG indiscriminately answers 100% of queries, hallucinating answers on 8 unanswerable items. DOCUTRUST routes ungrounded queries to human review or safe abstention.

Gold Class	DOCUTRUST (SYS-FULL)			Standard Baseline (BASE-C)		
	ANS	ESC	ABST	ANS	ESC	ABST
ANSWER ($N = 20$)	8	11	1	20	0	0
ESCALATE ($N = 6$)	0	4	2	6	0	0
ABSTAIN ($N = 8$)	0	5	3	8	0	0

keyword heuristics with typed `PolicyClause` rules; (6) corrected normal approximations with exact Student's $t(df = 25)$ and exact binomial McNemar calculations; (7) logged cryptographic provenance manifests.

VI. LIMITATIONS & FUTURE WORK

a) Elevated False Abstention Rate: While achieving a 100.0% Safe Non-Answer Rate, DOCUTRUST's 53.8% False Abstention Rate reflects high operational risk aversion. Future work will explore Bayesian calibration and conformal prediction to increase automated throughput.

b) Benchmark Expansion: The DOCUTRUST-BENCH pilot ($N = 50$) provides clean domain isolation but limited power for ranking significance. We are expanding the collection to $N = 400$ across ten enterprise verticals.

VII. CONCLUSION

We presented DOCUTRUST, an evidence-governed multi-agent architecture addressing the four failure modes of conventional RAG in compliance question answering: temporal supersession, authority hierarchy blindness, ungrounded hallucinations, and missing audit receipts. Supported by an extensive PRISMA review ($N = 66$ screened, $N = 56$ primary studies), DOCUTRUST achieves a 100.0% safe non-answer rate on unanswerable inquiries, an 85.7% conflict F1, a 91.7% claim support rate, and a steady-state median latency of 236.0 ms, backed by an independent audit and cryptographic SHA-256 provenance receipts.

REPRODUCIBILITY STATEMENT

All code, datasets, evaluation scripts, raw inference logs (408 records), independent audit files, and figure generation tools are fully open-source in the `research/` directory.

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